

CHAPTER

63 The Development of Attention

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Abstract

This chapter reviews past research and theory related to the early development of attention beginning with the most commonly used behavioral measures of infant visual attention. The developmental course of visual attention in infancy and early childhood is then described with a focus on the relevant behavioral and psychophysiological findings and recent work relating these findings to developmental change occurring in the brain in early development. Several theoretical models of the development of visual attention are discussed, including Richards' (2001, 2008, 2010) model of a general arousal/attention system. We conclude with a discussion of individual differences in infant attention and present a series of suggestions for future directions in research on attention in infancy and early childhood.

Keywords: [infancy](#), [visual attention](#), [event-related potentials](#), [cognitive development](#), [source analysis](#)**Subject:** [Cognitive Psychology](#), [Psychology](#)**Series:** [Oxford Library of Psychology](#)

Attention is a core component of adaptive responsiveness to the environment, and it is integral to all explicit cognitive processing. However, a clear and concise definition of attention as a global construct remains somewhat elusive (Colombo, 2001; Moray, 1969, 1993). It is generally agreed that there are multiple forms of attention and several component processes involved in attention. In one way or another nearly all of these forms of attention relate to focusing perceptual processing toward a particular aspect of the environment. Much developmental research has focused on the ability of infants and children of different ages to engage in selective attention, including orienting toward a specific stimulus and subsequently maintaining attention toward that central stimulus over distracters in the periphery. Rapid early developments in these aspects of attention are evident in a wide range of behaviors and also in the underlying brain structures that mediate them. Ultimately, they will have a significant impact on whether and how infants and young children will attend to people and events in their environment.

In this chapter we review past research and theory related to the early development of attention. We begin with a brief description of the most commonly used behavioral measures of infant visual attention. We then describe the developmental course of visual attention in infancy and early childhood. We review the relevant behavioral and psychophysiological findings and subsequently cover recent work relating these findings to developmental change occurring in the brain in early development. Additionally, we present Richards' (2001, 2008, 2010) model of a general arousal/attention system that provides a coherent and biologically plausible framework for understanding developmental changes that have been observed in attention across the first 2 years of postnatal life. We describe the impact of developmental change in Richards' general arousal/attention system on development of specific attention systems related to eye-movement control.

Historical Background and Behavioral Measures

Developmental research on attention in infancy has a relatively recent history. In the early years, the study of attention was of secondary interest in research where processes such as perception, discrimination, learning, strategy acquisition, planning, or working memory were of primary concern. Measuring any of these processes in early development is not a straightforward process because the human infant is incapable of making verbal responses or engaging in complex behavioral responses. Historically, researchers have thus utilized infant looking behavior as a window into attention and cognitive processing in infancy. Fantz (1964) was among the first to document a systematic decline in looking toward a stimulus with repeated presentations (i.e., habituation). In the habituation procedure, infants are exposed to repeated presentations of a single stimulus; once looking drops to a set criteria from the initial (or peak) looks (i.e., habituates), the stimulus is changed and subsequent looking to the new stimulus is examined. If the infant looks longer after the change has occurred, then the assumption is made that the infant recognized the difference between the new stimulus and the stimulus he or she was habituated to. This increase in looking following a stimulus change is often referred to as dishabituation or recovery of looking.

The paired comparison task is another procedure commonly used to examine infant looking behavior. In this procedure infants are shown a single stimulus during a brief familiarization phase. Following a delay interval, there is a recognition memory test in which the familiar stimulus is paired with a novel stimulus that has not been seen previously. Recognition memory for the familiar stimulus is inferred if the infants show greater attention (i.e., longer looking) to the novel stimulus than to the familiar stimulus during the paired comparison test. This procedure can also be applied without a familiarization phase to test the relative salience of stimuli based on previous experience outside of the laboratory. These looking time methods have provided a wealth of data on the conditions under which infants encode, store, and retrieve information. These include the facts that (1) encoding, storage, and retrieval processes in preverbal children could be examined objectively and scientifically; (2) even very young infants were able to recognize stimuli that they had seen before and in certain conditions could retain that information over several weeks; and (3) many of the processes and variables that were known to affect encoding, storage, and retrieval in older children and adults were integral to infant memory processes as well (for reviews, see Haith & Benson, 1998; Kellman & Banks, 1998; Mandler, 1992; Rose, Feldman, & Jankowski, 2004).

More recently, the study of attention has become important in its own right, as cognitive developmental research has become increasingly integrated with research in cognitive neuroscience. A number of practical issues have also contributed to the growing interest in the development of attention. For example, individual differences in attention observed during infancy are predictive of achievements in language, cognition, and play later in childhood. In addition, a number of deficits in attention processes contribute to or are symptomatic of the learning and behavior difficulties experienced by children with conditions such as attention-deficit/hyperactivity disorder (ADHD), fetal alcohol spectrum disorder (FASD), autism, and early exposure to teratogenic agents and environmental contaminants.

One focus that has emerged from the abundant data available from habituation and preferential looking experiments is the development of look duration in infancy. Changes in look duration occurring across infancy and early childhood are believed to reflect further development of areas of the brain involved in the control of eye movements and visual attention. Thus, the timing of change in visual behavior provides insight into the emergence of functional attention systems in the brain. Several models have been proposed to explain the development of visual attention from this perspective (e.g., Bronson, 1974, 1997; Hood, 1995; Johnson, 1990; Maurer & Lewis, 1979, 1998; Posner & Peterson, 1998; Richards, 2008, 2010; Richards, Reynolds, & Courage, 2010; Ruff & Rothbart, 1996). Many of these models were influenced by Schiller's (1985) work on primate eye-movement systems. In the next section we discuss the development of three specific attention systems: the reflexive system, the posterior orienting system, and the anterior attention system (or executive attention system). These systems involve networks of brain areas dedicated to

p. 1002 particular aspects or forms of attention. We also discuss a general arousal/attention system that influences all of these specific attention systems and also plays a key role in the developmental of attention.

The Reflexive System

In the newborn period, it is generally agreed that visual attention is primarily under the control of a reflexive system involving the lateral geniculate nucleus, primary visual cortex, and superior colliculus. Yet, in spite of the immaturity in both structure (e.g., retino-geniculo-cortical pathway) and function (e.g., spatial vision, color vision) of the human visual system (e.g., Atkinson, 2000; Banks & Salapatek, 1983; Hickey & Peduzzi, 1987), newborn infants do attend selectively to stimuli during their brief periods of alert inactivity. In his seminal research, Fantz (1963, 1964) showed that between birth and 2 months of age infants looked longer at patterned than at unpatterned stimuli and that depending on factors such as familiarity, size, and the amount of contrast, they preferred to look (i.e., looked longer) at some stimuli over others. Although visual reflexes (e.g., saccadic and pursuit movements) are present at birth, more general visuomotor immaturity restricts infants' ability to scan stimuli extensively or to detect stimuli beyond about 30 degrees in the peripheral visual field (e.g., Aslin, 1987; Lewis & Maurer, 1992). Moreover, when infants visually fixate on an element in the environment, they may have trouble disengaging from it, a phenomenon referred to as obligatory looking or "sticky fixation" (e.g., Hood, 1995). This is most evident between 1 and 2 months of age, a brief period when certain inhibitory mechanisms that limit eye movements come online. These early immaturities in the visual system do not last long. The rapid neurological development that occurs in the retina and in the visual pathways to the cortex between 2 and 3 postnatal months of age coincides with significant improvement in all aspects of visual functioning, an expansion of the effective visual field, moderation of inhibitory mechanisms that restricted eye movements, and the onset of more mature perceptual abilities whereby infants come to recognize objects and to determine their spatial layout. Coincident with this shift toward greater cortical control of vision, infants begin to spend more time awake and in an alert state. They look about the environment in a way that is less reflexive and more "voluntary" in nature (see Colombo, 2001; Haith, 1980).

Posterior Orienting System

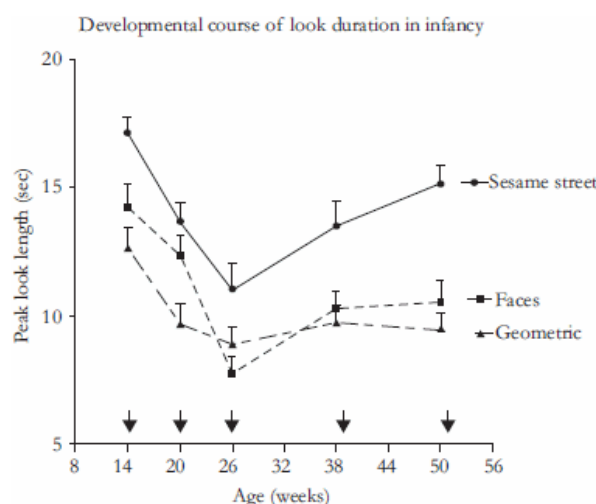
From 3 to 6 months of age, a posterior orienting network becomes functional (Posner & Peterson, 1990). Ruff and Rothbart (1996) described infants' visual behavior during much of the first postnatal year of life as dominated by an *orienting/investigative system* of attention. There are two interrelated components in this system. First, a spatial *orienting* network (including the posterior parietal cortex with several subcortical systems such as the superior colliculus, pulvinar, and the locus coeruleus in the brainstem), is alerted by peripheral stimuli and directs attention to potentially important locations in the environment (i.e., "where" an object is). This component mediates attention functions such as engagement, disengagement, shifting, and inhibition of return. Areas within the parietal cortex along with the frontal eye fields are integral for disengaging and shifting attention. As these areas develop, infants will be less prone to "sticky fixation" and develop greater ability to voluntarily shift attention. Second, an *object recognition* network (including the dorsal and ventral pathways from the primary visual cortex to the parietal cortex and the inferior temporal cortex) mediates attention to object features and gathers detailed information (e.g., form, color, pattern) that enables object identification (i.e., "what" an object is).

There is a marked developmental transition in the structure and function of this system between 3 and 9 months of age. Rapid maturation at all levels of the visual system along with increased periods of arousal and alertness enables infants to deploy their attention more flexibly and quickly and to respond to stimuli in terms of experiential factors such as the novelty or complexity of objects and events rather than by their intensity alone. As this attention system develops across this age range, infants will focus on the relevant features of objects and demonstrate clear visual preferences based on previous experience. The increased ability to shift attention combined with more efficient information processing leads to a linear decrease in looking from 3 to 6 months.

After 6 months of age, the rudiments of another major attention system, one in which the infant begins to acquire a system of higher level, endogenous, or voluntary control over the allocation and deployment of cognitive resources, begins to emerge. This capacity is evident in a wide range of behaviors. For example, infants' look duration to static stimuli and simple objects continues to decline, whereas their look duration to complex objects increases (Courage, Reynolds, & Richards, 2006; Ruff & Saltarelli, 1993). Infants also look more to their caregivers in situations that call for social referencing (Bertenthal & Campos, 1990) and joint attention (Bakeman & Adamson, 1984). At the same time, infants show the beginnings of behavioral inhibition on the A-not-B task (Diamond, 1985). Further evidence of emerging intentionality is evident in improvements in deferred imitation (Barr, Dowden, & Hayne, 1996; Barr & Hayne, 2000), means-end problem solving (Willatts & Rosie, 1989), and recall memory (Bauer, Wiebe, Waters, & Nelson, 2003) that occur late in the first year of life. By about 18 months of age, the endogenous control of attention acquires an increasingly executive function as toddlers also evaluate behavior progress and direct activity with goals and plans. The system involved in all of these developmental changes is called the anterior attention system because developments in executive attention and functioning are closely related to frontal brain activity, in particular the orbitofrontal cortex, dorsolateral prefrontal cortex, and anterior cingulate cortex (e.g., Bell & Fox, 1994; Chugani, 1994; Diamond & Goldman-Rakic, 1989; Posner, 1995). These changes in visual attention and their neural substrates enable (and may be enabled by) coincident changes in language (e.g., comprehension), cognition (e.g., representation), and self-regulation (e.g., behavioral inhibition) that begin in this time frame and continue to advance across the preschool years.

Colombo (2001) has proposed a triphasic theory of look duration that describes the overall trajectory of early developments in looking. This model fits well with the developing attention systems described earlier. The triphasic theory proposes that look duration increases from birth to 2 months of age, followed by a consistent decrease in looking from 3 to 6 months. After 6 months of age, there is a plateau in look duration. Courage, Reynolds, and Richards (2006) tested the triphasic theory in a developmental study of look duration in infants from 3 to 12 months of age. Infants were exposed to several different types of stimuli, including achromatic computer-generated patterns, a female face, and scenes from Sesame Street. Additionally, each stimulus type was presented in both static and dynamic conditions. Figure 63.1 shows the results of our analysis. The triphasic theory was supported overall with a clear drop in looking from 3 to 6 months of age (i.e., 14 to 26 weeks). However, after 6 months, the stimulus type began to have a significant impact

Figure 63.1



The developmental course of look duration to patterns, faces, and Sesame Street across infancy. The Y-axis represents infants' peak look lengths, and age (in weeks) is shown on the X-axis. Fourteen weeks corresponds to 3 months of age, 26 weeks corresponds to 6 months of age, and 52 weeks corresponds to 1 year of age. Arrows show exact test age. (Adapted with permission from Courage, M.L., Reynolds, G.D., & Richards, J.E. [2006]. Infants' visual attention to patterned stimuli: Developmental change from 3- to 12-months of age. *Child Development*, 77[3], 680–695.)

on look duration. Infants began to look longer to complex, interesting stimuli (faces, Sesame Street), whereas looking remained very low to basic, achromatic visual patterns. There was also a strong main effect

for dynamic stimuli, with infants looking longer to dynamic as opposed to static stimuli regardless of the specific type of stimulus (patterns, faces, Sesame Street). These data provide a clear picture of the development of visual attention systems and their impact on infant looking. The drop from 3 to 6 months in looking represents the onset of functioning of the posterior orienting system. The infant becomes able to shift fixation voluntarily from one stimulus to another, and this shifting becomes more rapid with increasing age. However, after 6 months, the anterior attention system begins to influence looking. Thus, looking remains brief to stimuli that require no further processing. But in situations where sustained attention is called for (dynamic patterns, faces, Sesame Street), the infant demonstrates increased looking and resistance to distracters. In addition to these three specific attention systems, there is a more general system that impacts attention throughout development.

General Arousal/Attention System

p. 1004 Attention can be seen as having two major functions: a selective function and an arousal function. ↪ The specific attention systems discussed earlier all relate to different aspects of the selective function in which attention determines what aspect of the environment the organism is focused on. The arousal function involves changes in arousal that are associated with attention. Ruff and Rothbart (1996) refer to this conceptualization of attention as “attention as state.” The changes in state associated with attention can be seen as fostering an optimal state of arousal for performance or learning. This arousal response is general in that it occurs across multiple modalities, is a component of all attentional responding (orienting, selection, and maintenance), and impacts functioning in both the central and peripheral nervous systems. Richards (2001, 2008, 2010) proposes that there is a general arousal/attention system in the brain composed of neuroanatomical connections between the mesencephalic reticular activation system and the cortex (Heilman, Watson, Valenstein, & Goldberg, 1987; Mesulam, 1983). Additionally, the noradrenergic and cholinergic neurotransmitter systems are highly involved in this system (Robbins & Everitt, 1995). This system shows substantial growth across early development with infants and children showing greater magnitude arousal responses associated with attention and children spending longer periods within this arousal state across age (e.g., Richards & Cronise, 2000).

Research over the past quarter of a century has been characterized by a steady increase in the sophistication of the tools available for measuring the arousal response associated with attention. Earlier work was aimed at establishing psychophysiological measures of infant attention and integrating these measures with existing behavioral measures of infant attention. Heart rate is the most notable of these psychophysiological measures and provides highly informative data regarding the process of attention in infancy. More recent work has established a link between behavioral and heart rate measures of attention and neural correlates of infant attention. Richards refers to times when the infant is attentive and the general arousal/attention system is engaged as “sustained attention.”

Sustained Attention

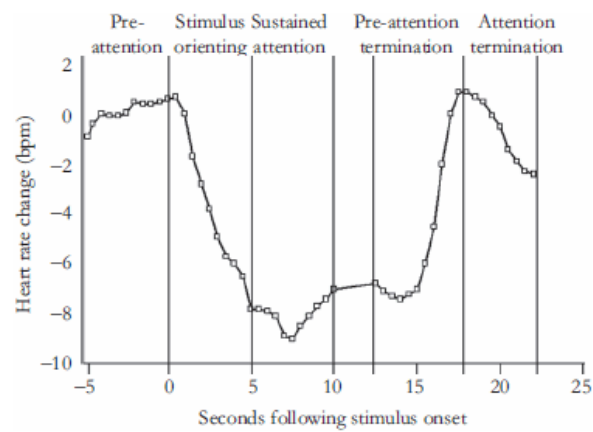
Sustained attention to stimulus features, also called focused attention, is the extended selective engagement of a behavior system that primarily enhances information processing in that system (e.g., see Richards, 2003). Many of the infant's cognitive and social activities occur during episodes of sustained attention, and it is during this time that active information processing occurs (see Oakes & Tellinghuisen, 1994; Reynolds & Richards, 2008; Ruff & Rothbart, 1996). Novel objects and events typically first elicit stimulus orienting followed by sustained attention. Maintenance of sustained attention facilitates the infant's learning and memory for aspects of the stimulus as it becomes familiar (Colombo, Richman, Shaddy, Greenhoot, & Maikranz, 2001; Frick & Richards, 2001; Richards, 1997a, 2003a). Infants are also less distractible by a peripheral stimulus and slower to orient to it when they are in sustained attention (Hicks & Richards, 1998; Hunter & Richards, 2003; Richards, 1997b; Ruff & Capozzoli, 2003). Infants as young as 3 months of age will engage in 5- to 10-second periods of sustained attention, and the duration that sustained attention can be maintained increases markedly to several minutes or more over the first 2 years of life (Reynolds & Richards, 2008; Ruff & Capozzoli, 2003). The development of sustained attention during the period of infancy is closely related to the development of brain systems controlling arousal and state (see Richards, 2001, 2008, 2010). These include the neuroanatomical connections between the mesencephalic reticular activating system in the brainstem, the thalamus, and the cortex. The mesencephalic reticular activating system also stimulates extrinsic neurotransmitters that in turn influence the limbic system areas. Sustained attention is a manifestation of this global arousal system of the brain that controls responsiveness to events in the environment. Sustained attention has been operationalized using both behavioral (i.e., looking) measures and heart rate, and we discuss details associated with each approach in the following sections.

Sustained Attention: Looking

The duration and direction of looking have been central measures of infants' attention to a variety of stimuli and over the years have provided important information about infants' sensory, perceptual, and cognitive processes and how they change with age. However, it is now evident that attention and looking are not isomorphic—there are varieties of attention and varieties of looking (see Colombo, 2001). Ruff and colleagues (Ruff, 1986; Ruff & Capozzoli, 2003; Ruff, Capozzoli, & Saltarelli, 1996; Ruff, Capozzoli, & Weissberg, 1998) identified and investigated two different patterns of looking that infants and young children display when given a novel toy to play with. An infant or child in “focused” attention showed a characteristic pattern of examination of the toy in which he or she looked at it with an intense facial expression and knit brows, in conjunction with reduced body movement, and manual rotation and finger examination of the toy. In contrast, during periods of “casual” attention, the infant continued to look at the toy but without the intensity of expression and the manual exploration of the toy. During casual attention the infant often waved or shook the toy, appearing to enjoy its sensory qualities (see also Oakes & Tellinghausen, 1994). Information processing occurs during focused attention but not in casual attention (see Oakes & Tellinghuisen, 1994; Ruff et al., 1996; Tellinghuisen & Oakes, 1997).

Sustained Attention: Heart Rate

Sustained attention is also indicated when infants are presented with a novel stimulus and an extended deceleration in heart rate accompanies looking at the stimulus. Richards and colleagues (Richards & Casey, 1992) proposed a model in which the phasic changes in infants' heart rates that occur as they look at a stimulus correspond to different levels of attentional engagement. There are four key phases in this model illustrated in Figure 63.2. First, when the stimulus is initially presented, there is a brief automatic interrupt phase that reflects the detection of a change in environmental stimulation. A very brief, reflexive, biphasic deceleration–acceleration in heart rate occurs. The second phase is stimulus orienting. Stimulus orienting indicates the beginning of attentional engagement and initiates preliminary



Richards' heart rate phases of attention. The average heart rate change as a function of seconds following stimulus onset during the heart rate-defined attention phases is shown (data from Richards & Casey, 1992). Change in heart rate from prestimulus levels is indicated on the X-axis, and time following stimulus onset is shown on the Y-axis.

processing of the information in the stimulus. A large, rapid deceleration in heart rate from its prestimulus level occurs during this phase. The third phase is sustained attention. Sustained attention reflects the activation of the alertness/arousal system of the brain and involves voluntary, subject-controlled cognitive processing of the stimulus information. The heart rate deceleration that was reached during stimulus orienting is maintained during this phase. Heart rate also shows decreased variability, and certain other somatic changes that facilitate attentiveness such as reduced body movement and slower respiration may also occur. Infants require less time to process and subsequently recognize a stimulus if exposure occurs during sustained attention than if exposure occurs during other heart rate phases (Richards, 1997a). The fourth phase is attention termination. During this last phase the infant continues to look at the stimulus but is no longer processing its information (i.e., is inattentive) and heart rate begins to return to prestimulus levels. Infants fail to demonstrate evidence of recognition memory for a previously viewed stimulus if exposure occurs during attention termination (Richards, 1997a).

The behavioral (looking) and psychophysiological (heart rate) indices of sustained attention are highly correlated (Lansink & Richards, 1997). The implications for information processing of focused and casual attention are similar to those of sustained attention and attention termination. Focused attention and casual attention are behaviorally defined periods of attention and inattention. Sustained attention and attention termination are heart rate-defined periods of attention and inattention. For example, infants take longer to disengage from a central stimulus and shift attention to a distracter stimulus during focused attention than during casual attention (Oakes & Tellinghuisen, 1994; Ruff et al., 1996; Tellinghuisen & Oakes, 1997). Similarly, infants demonstrate longer distraction latency to a peripheral stimulus presented during sustained attention than during attention termination (Casey & Richards, 1988; Hunter & Richards, 2003; Richards, 1987, 1997b). Sustained attention and focused attention are operationally defined phases of attention that represent periods when the infant is actively engaged in information processing. Distraction latencies are longer when infants are engaged in these phases of attention because infants are most likely processing information provided by a central stimulus. During casual attention or attention termination, infants respond more rapidly to peripheral stimuli because they are no longer engaged in attention to the central stimulus.

Heart rate and behavioral measures of infant attention also correlate with measures of brain activity associated with attention and recognition memory. Event-related potentials (ERPs) have been used to investigate the neural correlates of infant visual attention and recognition memory (e.g., Nelson & Collins, 1991, 1992; Reynolds, Courage, & Richards, 2010; Reynolds & Richards, 2005; Richards, 2003a,b). ERPs refer to voltage oscillations in the electroencephalogram (EEG) that are time-locked with an event of interest (Fabiani, Gratton, & Coles, 2000; Picton et al., 2000). Stimuli are presented briefly and repeatedly to each participant, and then trials are averaged together by experimental condition to identify the ERP. This averaging technique increases the signal-to-noise ratio in the EEG so that waveform components associated with a particular stage of processing can be identified. Of particular interest is the Negative central (Nc) component. The Nc is a component of negative polarity located over frontal and central electrodes with a peak latency of between 400 and 800 ms following stimulus onset. This component is ubiquitous in research on infant attention and memory and has been proposed to reflect novelty detection,

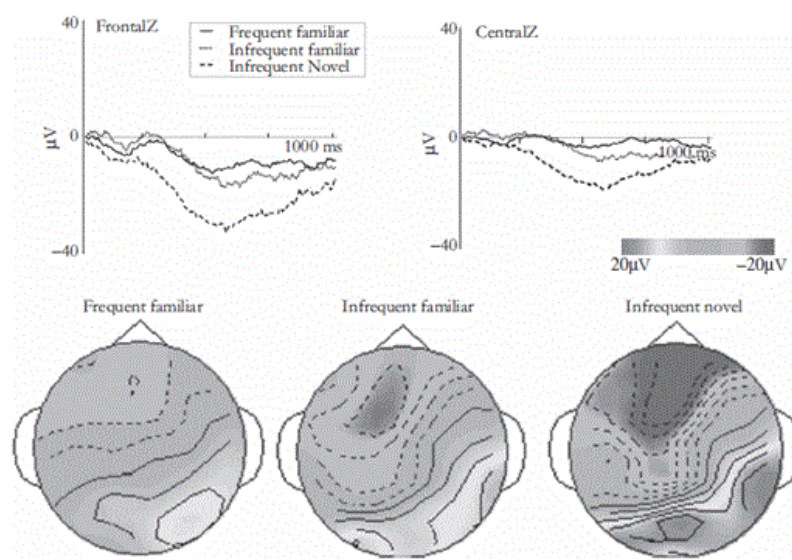
general orienting, or amount of attentional engagement (e.g., Carver, Bauer, & Nelson, 2000; Courchesne, 1977; Courchesne, Ganz, & Norcia, 1981; de Haan & Nelson, 1997, 1999; Karrer & Ackles, 1987; 1988; Karrer & Monti, 1995; Nikkel & Karrer, 1994; Reynolds & Richards, 2005, 2009; Richards, 2003a; Webb, Long, & Nelson, 2005).

Richards (2003a) explored the relationship between the Nc component and attention processes in a modified-oddball procedure that exposes infants to frequent familiar, infrequent familiar, and infrequent novel stimuli (Nelson & Collins, 1991, 1992). Infants of 4.5, 6, or 7.5 months of age were presented with a video of a Sesame Street movie as a background stimulus. Heart rate changes elicited by the video were used to distinguish periods of time during sustained attentiveness (during heart rate deceleration) and during inattentiveness (when heart rate was at prestimulus levels) (e.g., Casey & Richards, 1988; Reynolds & Richards, 2008; Richards, 1997, 2001; Richards & Casey, 1992). Nc amplitude was greater during periods of attention than inattention and also increased in magnitude across age, but only during periods of attentiveness. Reynolds and Richards (2005) used a high-density 128-channel EEG recording system that enabled the application of cortical source analysis of the ERP data for identification of locations in the cortex that could be potential generators of Nc (Reynolds & Richards, 2009; Richards, 2003b, 2004, 2005). Nc was found to be greater in amplitude to novel stimuli than familiar stimuli (see Fig. 63.3). Figure 63.4 shows the results of the source analysis displayed on a series of magnetic resonance imaging (MRI) images. The results of the cortical source analyses identified areas of prefrontal cortex, including the anterior cingulate, as likely sources of the Nc component.

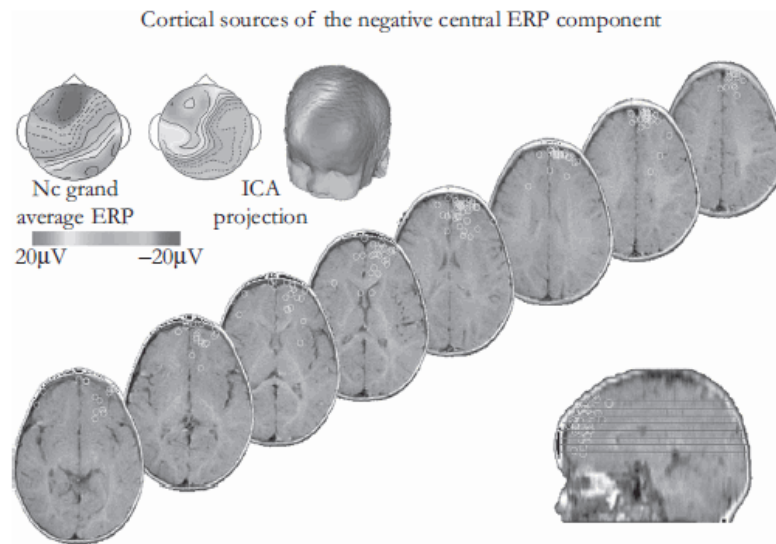
To determine the relationship between ERP and behavioral correlates of infant attention and memory, Reynolds, Courage, and Richards (2010) designed a procedure with alternating blocks of ERP trials followed by paired-comparison choice trials. This design allowed for the analysis of the distribution of infant visual preferences throughout ERP testing. There were several findings of interest. First, there was a main effect for age with Nc increasing in amplitude from 4.5 to 7.5 months of age. This may reflect the increasing involvement of prefrontal cortical areas in visual attention across infancy. Second, there was a main effect for attention replicating Richards (2003) with Nc being greater in amplitude during sustained attention. Third, there was an interaction of preference with stimulus type. Infants that demonstrated novelty preferences demonstrated greater amplitude Nc in response to novel versus familiar stimuli. The opposite pattern was found for those that demonstrated familiarity preferences.

These findings indicate that Nc is impacted by stimulus salience. Thus, Nc reflects activation of the general arousal system involved in attention. Through the use of heart rate, ERPs, behavioral measures, and source localization, different levels of this general arousal system governing attention can be examined. Activation of this system leads to decreased heart rate through the parasympathetic influence of the brainstem on the heart. The general arousal system also involves enhanced processing throughout the cortex through the influence of the noradrenergic and cholinergic neurochemical systems. Thus, areas of the brain involved in the specific attention systems will demonstrate enhanced activity when the general arousal/attention system is engaged. Our results support this proposal showing that areas of the cortex that demonstrate enhanced activation when this attention system was engaged include inferior and superior prefrontal cortex, and the anterior cingulate cortex (all structures involved in the anterior attention system).

Neuroimaging work with adult participants has shown that these areas are active in a wide range of tasks involving attention (see review, Duncan & Owen, 2000). Thus, with advances in technology and increased sophistication of tools available to developmental scientists, a clearer picture has emerged of the normal course of early development of attention and the role that brain development plays in this process. An important question that remains to be answered concerns the specific role of the brain in individual differences in infant attention.

Figure 63.3

The Nc component following frequent familiar, infrequent familiar, and infrequent novel stimulus presentations. The event-related potential waveforms are shown by stimulus type in the top panel at FrontalZ and CentralZ electrode locations (see Reynolds & Richards, 2005). The bottom panel shows the topographical scalp potential maps of the distribution of this component for the three stimulus types at the peak of Nc (approximately 500 ms following stimulus onset). (See color insert.)

Figure 63.4

The sequence of magnetic resonance imaging (MRI) slides shows the dipole locations (yellow circles) for an event-related potential (ERP) component known as the Negative central (Nc) (topographical scalp potential maps on upper left figures). The dipoles represent the best-fitting cortical sources of the Nc for individual participants obtained through cortical source analysis of high-density ERP data. (See color insert.)

Because performance on the habituation and paired-comparison procedures has been found to have significant predictive value for cognitive functioning in childhood (e.g., Bornstein & Sigman, 1986; Fagan, 1984; Rose & Feldman, 1995), the study of individual differences in infant cognition has emerged as an important area of research. The best known approach to studying individual differences in infant visual attention is to examine differences across infants in their look durations during habituation to or during familiarization with a novel stimulus. Using this approach, look duration has been found to be negatively correlated with measures of cognitive performance in infancy as well as certain measures of intellectual performance obtained in later childhood years (Bornstein & Sigman, 1986; Colombo, 1993; Colombo & Mitchell, 1990; Colombo, Mitchell, Coldren, & Freeseaman, 1991; Colombo, Mitchell, & Horowitz, 1988; McCall & Carriger, 1993; Rose, Slater, & Perry, 1986; Sigman, Cohen, Beckwith, & Parmalee, 1986; Tamis-LeMonda & Bornstein, 1989). According to Colombo and Mitchell (1990), peak look-length during habituation or familiarization provides the most reliable and stable indicator of individual differences in attention and cognitive processing. Infants that demonstrate brief visual fixations (i.e., short-lookers) are more likely to demonstrate evidence of recognition memory for a familiar stimulus than infants that demonstrate long visual fixations (i.e., long-lookers).

Mechanisms that have been proposed to account for individual differences in infant visual attention include speed of processing, processing strategy, and the ability to disengage attention (Colombo, 1993; Colombo, Freeseaman, Coldren, & Frick, 1995; Colombo et al., 2001). Colombo and colleagues (2001) tested speed of processing and processing strategy with 4-month-old infants using the paired-comparison procedure, which consisted of the simultaneous presentation of two novel stimuli that differed from the familiar on either global or local stimulus properties. Familiarization time was also manipulated. Results indicate that with brief familiarization time (20 s), short-lookers only demonstrated preferences (i.e., looked longer) based on global differences; long-lookers failed to demonstrate a preference for either condition. With longer familiarization times (30s), short-lookers shifted from demonstrating global-differences preferences to demonstrating local-differences preferences. Long-lookers required the longest familiarization time (50s) to demonstrate a preference, and this only occurred to local differences. The authors concluded that (1) short-lookers demonstrate a dominant tendency for processing the global properties prior to the local properties of a visual stimulus and that (2) long-lookers process information at a slower rate than short-lookers and use a less efficient visual intake strategy by processing the local properties prior to the global ones.

Colombo, Richman, Shaddy, Greenhoot, and Maikranz (2001) found that long-lookers spend proportionately more time in attention termination than short-lookers. This indicates that long-lookers may have problems disengaging fixation. Long-lookers are more likely to continue looking at a stimulus while inattentive, whereas short-lookers are more likely to shift fixation away from a stimulus once attention has waned. Thus, the addition of heart rate measures to the video analysis of looking behavior provided a much more accurate measure of infant attention and provided a level of insight into possible mechanisms behind individual differences in infant attention that was not possible with looking behavior alone.

Reynolds, Guy, and Zhang (2011) conducted an investigation of individual differences in infant visual attention and ERPs. Infants of 6 and 7.5 months of age were tested in a procedure in which they were familiarized with a single stimulus, and then shown the familiar stimulus on 50% of the ERP trials, and novel stimuli on the other 50% of trials. Look duration during familiarization was used to determine looker type. There were a couple of interesting findings. There was a significant interaction of looker type and stimulus type on the late slow wave at right temporal electrodes. The late slow wave is commonly found to occur on novel trials and decreases in amplitude with repeated exposure (e.g., de Haan & Nelson, 1997; Quinn, Westerlund, & Nelson, 2006; Snyder, 2010). Short-lookers demonstrated a late slow wave to novel stimuli and a return to baseline to the familiar stimulus. The return to baseline following familiar presentations indicates the short-lookers fully processed and recognized the familiar stimulus. Long-lookers demonstrated no differences in late slow waves to both novel and familiar stimuli, indicating incomplete processing and lack of recognition of the familiar stimulus. These findings demonstrate that individual differences in visual attention are related to differences in cortical activity across infants.

There is a rich history of research on the early development of visual attention. Early work in the area capitalized on the natural tendency of human infants to look longer at novel compared to familiar stimuli. Habituation and preferential looking measures have provided a wealth of information regarding the developmental course of looking and visual attention in infancy and early childhood (for reviews, see Colombo, 2001; Haith & Benson, 1998; Keller & Banks, 1998; Mandler, 1992; Rose, 2007). Models of the development of visual attention and the underlying changes in the brain responsible for this development have been traditionally based on behavioral data obtained through looking measures with human infants, integrated with neuroanatomical data obtained through comparative animal research (e.g., Bachevalier, 2008; Schiller, 1985). This application of comparative data to explain developmental change in human infants has been useful but has certain limitations (see Richards, 2010 for a detailed discussion). More recent work has utilized noninvasive measures of infant brain activity (e.g., EEG, ERP, structural MRI, source analysis) to test these models directly in human participants (e.g., Reynolds & Richards, 2005; Richards, 2001, 2005, 2007).

The extant findings provide support for the proposal that there are several attention systems that show substantial development in infancy. Most notable are the posterior orienting system, anterior attention system, and a general arousal/attention system (see Ruff & Rothbart, 1996; Posner & Peterson, 1990; Richards, 2001, 2008, 2010). As these systems become functional, visual attention and looking show clear developmental change. Early attention is primarily under the control of a reflexive system that is characterized by obligatory attention to salient stimuli and motion in the periphery and also is highly susceptible to sticky fixation. As the posterior attention system develops, areas of posterior parietal cortex (PG), and the frontal eye fields become involved in voluntary shifts of attention. This system shows substantial development across 3 to 6 months of age, a time when visual attention is characterized by increasingly shorter looking. From 6 months on, the anterior attention system becomes functional, and frontal areas (inferior and superior prefrontal cortex, anterior cingulate) begin to play a role in sustained attention toward stimuli when attention and information processing is required. Attention to distracters is inhibited in these cases. Thus, visual attention and looking become increasingly voluntary and functional after 6 months of age, representing early examples of “executive attention.”

The general arousal/attention system influences all three of these more specific attention systems. As the child develops, the ability to maintain an aroused/attentive state increases. During activation of the general arousal/attention system, activity of the specific attention systems in the brain is enhanced through a generalized spread of arousal (decreased heart rate, activation of the noradrenergic and cholinergic neurochemical systems). Alternatively, as the anterior attention system becomes functional, the child develops the ability to voluntarily maintain attention toward a central stimulus while inhibiting attention to distracters. This voluntary control of attention allows the child to actively maintain sustained attention—the primary component of the general arousal/attention response. This rudimentary form of executive attention will be critical for the development of more complex behaviors and cognitive processes (e.g., language, self-regulation, mental representation).

Future Directions

1. Further refinement is needed for applying cortical source analysis to infant EEG/ERP data. The existing models for estimating the cortical sources of EEG activity are based on adult parameters (skull/scalp thickness, etc.). Richards (see Reynolds & Richards, 2009) is currently developing a technique that uses structural MRIs from individual infant participants to model the brain activity that produces their EEG data based on the anatomical features of their own brain as opposed to the typical anatomy of the adult brain. Ultimately, the ability to apply a direct measure of brain activity (e.g., fMRI) with infant participants is needed; however, practical and ethical concerns currently preclude the use of fMRI with normal infant participants. Structural MRIs are typically obtained from infants while sleeping.

2. The research described in this chapter is based largely on studies of attention processes in normally developing infants and children. Little is known about the development of attention in atypical and clinical populations. Once progress had been made with this, subsequent work could enable the identification of infants who are at risk for developmental disorders. Ultimately, identifying early markers for children at

risk for developing attention-related disorders or learning disabilities will facilitate early intervention and potentially better outcomes.

p. 1013 **Further Reading**

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